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Experiment No.: 02

Title: Implementation of Linux Shell Scripts Using grep, egrep, awk, and System Commands.

Objective:

- To write shell scripts using grep and egrep for file and directory operations.
- To develop awk scripts for generating Fibonacci series and displaying patterns.
- To write shell scripts to retrieve system information such as OS details, process status, memory usage, and user environment variables.

Software: Linux operating system

Theory: Shell scripting in Linux allows automation of system tasks using command-line utilities. Tools like grep and egrep are used for pattern matching, while awk is a powerful text-processing language. System commands enable users to access operating system details, process information, and environment settings efficiently.

Output/Result:

```bash
#!/bin/bash

if [ $# -ne 1 ]; then
    echo "Usage: $0 filename"
    exit 1
fi

file=$1

echo "File Statistics for: $file"
echo "----------------------------"

# Count lines
lines=$(grep -c "" "$file")

# Count words
words=$(grep -oE '\w+' "$file" | wc -l)

# Count characters
chars=$(grep -o "." "$file" | wc -l)

echo "Lines      : $lines"
echo "Words      : $words"
echo "Characters : $chars"
```

```
siesgst@siesgst-OptiPlex-3000: ~/oslab2/
Linux is an open source operating system.
It is powerful, flexible, and widely used.
Operating Systems manage processes, memory, and files.
Shell scripting makes system administration easier.
Awk, grep, and sed are powerful text processing tools.
```

```
File Statistics for: file.txt
-----------------------------
Lines      : 6
Words      : 36
Characters : 242
```

```bash
#!/bin/bash

read -p "Enter number of terms: " n

awk -v n="$n" '
BEGIN {
    a = 0
    b = 1
    for (i = 1; i <= n; i++) {
        printf "%d ", a
        c = a + b
        a = b
        b = c
    }
    printf "\n"
}'
```

```
#!/bin/bash

read -p "Enter a string: " str

awk -v s="$str" '
BEGIN {
    for (i = 1; i <= length(s); i++) {
        print substr(s, 1, i)
    }
}'
```

```
Enter a string: HELLO
H
HE
HEL
HELL
HELLO
```

```
#!/bin/bash
ls | egrep '.*'
```



```
22a.sh
22b.sh
22c.sh
22d.sh
22e.sh
Desktop
Documents
Downloads
list.sh
Music
nishank
oslab2
Pictures
Public
Templates
Videos
```

```
#!/bin/bash

echo "Operating System Information"
echo "----------------------------"

if [ -f /etc/os-release ]; then
    . /etc/os-release
    echo "OS Name        : $NAME"
    echo "OS Version     : $VERSION"
    echo "Release ID     : $VERSION_ID"
else
    echo "OS information : Not available"
fi

echo "Kernel Version : $(uname -r)"
```

```
Operating System Information
----------------------------
OS Name        : Linux Mint
OS Version     : 22.2 (Zara)
Release ID     : 22.2
Kernel Version : 6.14.0-29-generic
```

```
#!/bin/bash
ps -eo pid,ppid,cmd,%cpu,%mem --sort=-%cpu | head -n 11
```

```
    PID    PPID CMD                         %CPU %MEM
  76862    2114 /opt/google/chrome/chrome -  6.3  3.7
  74716    2921 /usr/lib/firefox/firefox-bi  3.1  3.3
  72636    2921 /usr/lib/firefox/firefox-bi  2.6  2.9
  80357       1 /usr/libexec/flatpak-system  2.4  0.3
  79619    2114 /opt/google/chrome/chrome -  1.3  2.3
   2092       1 /opt/google/chrome/chrome    1.1  4.2
   2175    2114 /opt/google/chrome/chrome -  1.0  2.0
   2476    1546 /usr/lib/firefox/firefox     0.8  6.2
  80024    1099 /usr/libexec/gnome-terminal  0.5  0.5
   3219    2921 /usr/lib/firefox/firefox-bi  0.5  3.2
```

```
#!/bin/bash
ps -eo pid,ppid,cmd,%mem --sort=-%mem | head -n 11
~
~
~
```

```
   PID    PPID CMD                          %MEM
  2476    1546 /usr/lib/firefox/firefox      6.3
  7617    2114 /opt/google/chrome/chrome -   5.8
  2092       1 /opt/google/chrome/chrome     4.2
 76862    2114 /opt/google/chrome/chrome -   4.0
 74716    2921 /usr/lib/firefox/firefox-bi   3.3
  3219    2921 /usr/lib/firefox/firefox-bi   3.2
 72636    2921 /usr/lib/firefox/firefox-bi   2.9
  7915    2114 /opt/google/chrome/chrome -   2.5
  2573    2114 /opt/google/chrome/chrome -   2.3
```

```
#!/bin/bash
echo "Current User: $(whoami)"
echo "Login Name : $(logname)"
~
~
~
~
~
```

```
#!/bin/bash
echo "Current Shell : $SHELL"
echo "Home Directory: $HOME"
echo "OS Type       : $(uname -o)"
echo "PATH Setting  : $PATH"
echo "Working Dir   : $(pwd)"
~
~
~
```

Conclusion: In this experiment, various Linux shell scripts were successfully implemented using grep, egrep, and awk. The scripts efficiently performed text processing, pattern matching, and system monitoring tasks. This experiment enhanced understanding of shell scripting and system-level automation in Linux.